

ANNE CHRISTIONO

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EDUCATION:

Massachusetts Institute of Technology, Cambridge, MA

August 2024 - May 2027

Bachelor of Science in Computer Science & Engineering and Mathematics

GPA: 5.0/5.0

Courses: Deep Learning (G), Machine Learning, Algorithms, Statistics, Probability & Random Variables, Linear Alg, Programming
Activities: Machine Learning L.A., Multivariable Calc Grader, MIT Hacking Medicine (Exec), Society of Women Engineers (Exec)

William P. Clements High School, Sugar Land, TX

May 2024

High School Diploma, Summa Cum Laude

GPA: 4.67/4.00 | Rank: 5/640 | SAT: 1580 (R: 780; M: 800) | AP: 5/5 on 14 exams

SKILLS:

Python, Java, C++, SQL, HTML/CSS, Postgres, SHAP, LangChain, JavaScript, MATLAB, Pytorch, Scikit-learn, Tensorflow, Git, Data Structures, Causal Graphs, Sentiment Analysis, Feature Selection, Model Explainability, Communication, Problem-Solving

PROFESSIONAL EXPERIENCE:

Software Engineering Intern, Microsystems Technology Laboratories

Cambridge, MA, USA

PostgreSQL Patent Database & Search Platform for Research and Industry Collaboration

03/2025 – present

- Integrating transformer-based LLMs with PostgreSQL for semantic search, query parsing, & automated patent summarization
- Developing full-stack system w/ Python backend, REST APIs, and HTML/CSS/JS frontend for scalable patent discovery

Human-AI Interaction Researcher, National University of Singapore (NUS) Computing

Singapore

Multimodal Computational Psychometric Modeling of Human Attitudes and Behavioral Intentions

03/2025 – 08/2025

- Built multimodal AI chatbot integrating LLMs with TTS/STT, enabling real-time sentiment detection in interviews
- Extracted text and voice features, applying ML-driven causal graph analysis to reveal key factors behind mental health stigma

Machine Learning Research Intern, Houston Methodist Research Institute

Houston, TX, USA

Predicting Nanoparticle Toxicity Through Machine Learning-Based Models

05/2023 – 06/2025

- Engineered and evaluated 18 tree-based, nonlinear, and linear predictive models (XGBoost, CatBoost, SVM, KNN) with Scikit-learn to classify nanoparticle toxicity using curated dataset of 8,000+ samples and 19 physicochemical features
- Implemented nested 10-fold cross-validation and GridSearch for hyperparameter optimization and robust model selection
- Utilized SHAP analysis for model explainability & feature selection; quantified generalizability (88.7% accuracy on new data)

AI/ML Undergraduate Researcher, MIT Media Lab

Cambridge, MA, USA

Harnessing Language Models for Autonomous Biological Discovery

09/2024 – 01/2025

- Developed PyLabRobot, versatile cross-platform Python interface to program wide range of liquid-handling Opentrons robots
- Leveraged LangChain (LLMs) to implement PyLabRobot features to execute experiments based on natural language prompts

Data Science Research Intern, The MITRE Corporation

McLean, VA, USA

State-of-the-Practice Analysis of U.S. Synthetic Biology Ecosystem

06/2022 – 08/2022

- Applied statistical modeling, data analysis, and market research on iGEM datasets, literature, and SME interviews in SynBio & Computational Biology for DoD-sponsored SynBio Moonshot, generating data-driven insights to guide R&D strategies
- Published internal reports and presented to MITRE leadership, influencing strategic direction in synthetic biology innovation

PUBLICATIONS:

- Cave, J., **Christiono, A.**, et al. (2025). Rational Design of Safer Inorganic Nanoparticles Via Mechanistic Modeling-informed Machine Learning. *ACS Nano*, 19(23), 21538–21555. <https://doi.org/10.1021/acsnano.5c03590>
- Christiono, A.**, et al. (2022). The Wiener Index: From Trees to Graphs with Many Cut-Edges [Conference presentation abstract]. 53rd Southeastern Int'l Conference in Combinatorics, Graph Theory, and Computing, Florida Atlantic University.

HONORS AND AWARDS:

- USA Computing Olympiad (USACO): Gold Division Competitor, 2021
- National Center for Women & Information Technology: 2x National Honorable Mention & Houston Award Winner, 2022-23
- American Invitational Mathematics Examinations (AIME): 5x Qualifier, 2020-2024
- Science Olympiad: 2x National Medalist, 2x National Science Olympiad Qualifier, 25+ medals in CS/math/biology events
- USA Biology Olympiad (USABO): Semifinalist, March 2023
- New York Times STEM Writing Contest: Runner-Up (Top 24 of 3,500+ submissions worldwide), May 2022
- Piano: 4x National Gold Cup by the National Federation of Music Clubs; 2022 Outstanding All-State Winner

LEADERSHIP AND COMMUNITY INITIATIVES:

NUS High School of Math and Science: Teacher for Advanced Calculus & Machine Learning

01/2025 – 02/2025

- Prepared and led calculus and ML courses for penultimate-year students as part of MIT's Global Teaching Labs in Singapore

INTEGIRLS Houston: Advisor (present), Chapter Director (2022-2024), Web Director (2021-2022)

08/2021 – present

Global youth-led non-profit to bridge the gender gap in STEM through free, virtual math contests for female & nonbinary students

- Led 9 team members to organize contests (participants doubled to 130+ globally); raised \$10K+ from sponsors annually (doubled from previous year, newly acquired sponsorship from AOPS, Hudson River Trading, Jane Street, Geogebra)